

NAME: **RAGHURAM BALAJI**

Designation: Senior Engineer

Emp Id: 132838

E-mail: raghuram.balaji@coforge.com

Mob. No.: +91 7989614165

Experience: 11+ years

Location: Hyderabad, India



PROFESSIONAL SUMMARY

- Systems and platform engineer with **11+ years** across **native C/C++** engine internals, **Java/Spring** backend services, and cross-platform debugging — currently embedded with Unity Technologies engineering teams on runtime and compiler-toolchain stability.
- Deep expertise in root-causing the hardest class of defects: **memory leaks**, GC/runtime anomalies (**Mono/IL2CPP**), **GPU/rendering-pipeline** issues, OS- and driver-specific regressions across Windows, Linux, Android, and WebGL targets.
- Strong Java backend engineering: **Java 8**, **Spring Boot/MVC** REST services, **WebSocket/STOMP** real-time workflows, PostgreSQL/SQL Server/Redis, deployed on self-managed Linux and cloud (OVHcloud, AWS, Azure).
- Proven record of regression-safe fixes and **architectural backports** across **LTS releases** of a large production codebase under strict backward-compatibility requirements.
- **Methodologies / Tools**: Agile (Scrum), Jira, Yamato (internal CI), Git, Mercurial (hg), Perforce, Unity VCS (Plastic SCM), Visual Studio, VS Code.
- Domain exposure spanning developer tools / real-time 3D (enterprise customers from automotive to gaming and kiosk systems), aerospace & heavy-vehicle manufacturing, media/music streaming, and Intel OEM consumer hardware.

TECHNICAL SUMMARY

Technologies/ Framework	• Systems: C, C++ (11/14), Mono runtime internals, IL2CPP, Emscripten/WebAssembly, OpenGL/GPU pipelines, Win32 API, DLL/ABI design • Backend: Java 8, Spring Boot, Spring MVC, REST, WebSocket/STOMP (SockJS), HLS (.m3u8), HTTP Range Requests • Web/Client: C#, React (Vite), JSON, WebRTC • Testing: NUnit (C#), UnitTest++ (C++), CI/CD & PR review workflows
Domain/Industries	Developer Tools / Real-Time 3D (automotive, gaming, kiosk verticals), Aerospace & Heavy-Vehicle Manufacturing, Media & Music Streaming, Consumer Hardware (Intel OEM)
Build Tools	Yamato (internal CI), CI/CD pipelines with automated test gates, Tomcat/Apache hosting, shell-scripted deployment
Databases	PostgreSQL, SQL Server, Redis
Version Control	Git, Mercurial (hg), Perforce, Unity VCS (Plastic SCM)
Loggers	Native crash-dump & trace analysis, engine/editor logs, Yamato/internal CI logs
IDE	Visual Studio, VS Code
Operating Systems	Windows, Linux (RHEL-compatible); Android & WebGL as debug targets
Software Methodologies	Agile (Scrum), Jira

CLIENT / PROJECT DETAILS

Company Name : Coforge Client Name : Unity Technologies (Unity3D UK / ISS)		Duration : 05/2025 to Present
Team Size	15	
Technologies	C/C++, Mono runtime internals, IL2CPP compiler, GC/memory-leak analysis, native profiling, crash-dump & trace analysis	
Role	Senior Engineer — Runtime Systems	
Project Overview	Deep-level stability engineering on the Mono scripting runtime and its tightly coupled IL2CPP compiler pipeline — diagnosing the hardest class of defects (memory leaks, GC anomalies, platform-dependent failures) at the native runtime level of a large-scale production engine.	
Roles & Responsibilities	- Root-cause memory leaks and runtime defects in the Mono scripting runtime, tightly coupled to the IL2CPP compiler pipeline — native C++ debugging of allocation paths, GC behavior, and generated-code interactions across platforms. - Diagnose platform-dependent failures (e.g., driver-specific defects where behavior diverged on one platform while identical code ran clean on others) via native traces, crash dumps, and source-level debugging.	

Company Name : Coforge (formerly Rythmos India Pvt. Ltd.) Client Name : Unity Technologies (Unity3D UK / ISS)		Duration : 07/2023 to 04/2025
Team Size	15	
Technologies	C++, C#/Mono, WebGL (IL2CPP/Emscripten/WebAssembly), OpenGL/GPU pipelines, Perforce, Unity VCS (Java-backed services), NUnit, UnitTest++, CI/CD	
Role	Senior Consultant — Core Engine & Platform Systems	
Project Overview	Core C++ engine/platform engineering across the editor (C#/Mono), the WebGL (IL2CPP/Emscripten) runtime, low-level graphics, and version-control integrations — with LTS backports and automated test coverage.	
Roles & Responsibilities	- Diagnosed and patched GPU dependencies, rendering-pipeline issues, and WebGL/OpenGL compilation defects; resolved OS-specific regressions where shared C# behavior diverged across Windows, Linux, and Android due to OS/hardware differences. - Owned ~1 year of Unity Version Control (UVCS)/Perforce plugin engineering deep in the engine's C++ source, integrated with UVCS's Java-backed services — e.g., preventing fetched/synced .asset files from wrongly entering the pending changelist and losing their read-only flag during re-serialization. - Root-caused WebGL Video Player defects (irregular playback speed, VideoPlayer.Play() frame delay) across the C#/Mono editor layer, server-side m3u8 streaming, and the IL2CPP > Emscripten > WebAssembly buffer/caching path — a module underpinning CDN-backed video delivery and ad-serving. - Executed architectural backports across LTS releases under strict backward compatibility; authored automated tests (NUnit/C#, UnitTest++/C++) integrated into CI/CD and PR review workflows.	

Company Name : Coforge (formerly Rythmos India Pvt. Ltd.) Client Name : Unity Technologies		Duration : 12/2021 to 06/2023
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Team Size	15
Technologies	Java 8, Spring Boot, REST, WebSocket/STOMP (SockJS), React (Vite), C++ + serialization layer, JSON, Linux (OVHcloud dedicated servers), AWS
Role	Senior Consultant — Java Backend & Native Data Layer
Project Overview	Localization platform connecting a distributed translation web application to editor tooling — Java backend rebuild plus native C++ serialization-layer integration, in collaboration with a remote backend team in Japan.
Roles & Responsibilities	• Rebuilt the legacy localization backend on Java 8/Spring Boot REST services (React front end retained), consumed by both the distributed translation platform and editor tooling; platform ran on OVHcloud dedicated Linux servers (EU data residency), with migration to AWS underway during the engagement. • Extended the translation web app (React + Vite SPA) with an event-driven submit > review > approve workflow — live updates pushed over WebSocket/STOMP (SockJS) simultaneously to the web UI and editor-side tooling without page refresh; approved changes synced back as structured JSON keyed by project/scene/asset/UI element. • Implemented the client-side integration at the native C++ serialization layer — modifying how localized variables/state were serialized and transferred so changes propagated automatically into the editor's data-bound UI, with no per-window UI code required.

Company Name : Rythmos India Pvt. Ltd. (now Coforge) Client Name : Unity Technologies (Unity3D SE Team)		Duration : 07/2019 to 12/2021
Team Size	15	
Technologies	C++, C#, native debugging, crash/trace analysis, LTS backports	
Role	Associate Consultant — Enterprise Support / Platform Stability	
Project Overview	Engineering-level support for a large-scale production engine serving enterprise customers across automotive, gaming/slot-machine, and kiosk (airport/hospitality) verticals — defect reproduction, root-cause analysis, and LTS backporting.	
Roles & Responsibilities	• Reproduced and root-caused enterprise customer defects in the engine via minimal repro projects, logs, crash dumps, and source-level native debugging; delivered fixes across the C#/C++ codebase. • Ported platform-side C++ bug fixes and backports to LTS releases, partnering with enterprise clients to reduce escalations.	

Company Name : iCube Systems / iCube Logics (I) Pvt. Ltd. Client Name : Confidential (Aerospace & Heavy-Vehicle Manufacturer — NDA)		Duration : 06/2017 to 05/2019
Team Size	5	
Technologies	Java 8, Spring MVC, REST, SQL Server (JDBC), Tomcat/Apache, RHEL-compatible Linux, Spring WebSocket/STOMP (SockJS), WebRTC, CNN-based vision pipeline, Azure, FTP/file-drop ingestion, legacy Windows interop (Win32/MFC, COM/DCOM, VB6, WCF/IIS)	
Role	Technical Lead — Simulation / Java Backend Lead	
Project Overview	Java 8/Spring MVC integration layer on Linux enabling warehouse-	

	automation modules to consume the enterprise SQL Server datastore shared across multiple verticals — while preserving untouched compatibility with design/engineering teams on Windows-based Siemens/CAD toolchains.
Roles & Responsibilities	- Architected a centralized REST API layer around legacy Windows systems (Win32/MFC, COM/DCOM, VB6 clients, WCF/IIS services, registry- and path-bound workflows), giving new modules one clean integration point without disturbing cross-vertical schemas, reports, or ODBC/OLE DB consumers. - Built Spring MVC services powering warehouse operations — part-dependency algorithms and REST APIs letting factory robots query QR-coded bins, calculate assembly capacity, and auto-trigger restocking; bridged robotic retrieval with the Java/Spring WMS, cutting supply-chain restocking delays by 85%. - Built an in-panel support system with real-time chat (Spring WebSocket/STOMP over SockJS) escalating to WebRTC peer-to-peer video calls for machine-breakdown support — STOMP topics brokered the WebRTC signaling (SDP/ICE exchange); kept in-house vs. third-party tools for data sensitivity. - Engineered a live video-frame pipeline feeding a CNN (semantic segmentation) for on-the-fly defect detection — built the streaming infrastructure ahead of the model so detection could be enabled via feature flag once trained. - Hosted services on Tomcat/Apache with FTP/file-drop pipelines for machine-input ingestion, inspection imagery, and simulation/automation data exchange.

Company Name : Vroovy Tech (Hungama Gameshastra Pvt. Ltd.) Client Name : Hungama Digital Media		Duration : 11/2015 to 06/2017
Team Size	15	
Technologies	Java, Spring, Linux, PostgreSQL, Redis, HLS (.m3u8), HTTP Range Requests, shell-scripted deployment	
Role	Junior Engineer (Java Backend)	
Project Overview	Java/Linux middleware service between client applications and Hungama media servers — centralizing playlist logic, music-metadata lookup, and HLS audio delivery in a thin-client architecture so business rules lived server-side instead of being hardcoded in apps.	
Roles & Responsibilities	- Built Spring-based backend services handling server-driven UX flows, state machines, and playlist/business rules; PostgreSQL as the system of record for music catalog metadata, media references, and playlist configuration, with Redis fronting low-latency transient state (sessions, playback flow, hot lookup data). - Engineered HLS pipeline logic in Java — fetching and parsing .m3u8 manifests, resolving audio segments, and applying server-side randomization/selection rules — then streaming selected chunks via HTTP Range Requests (partial content) directly from Hungama media servers, eliminating full-file downloads. - Owned Linux deployment workflow: JVM runtime tuning via environment properties, shell-scripted rollout, and service configuration across environments.	

Company Name : virtualinfocom Client Name : Intel (sponsored project)		Duration : 06/2014 to 05/2015
Team Size	4	
Technologies	C, C++11, Win32 API, DLL/ABI design, extern "C" exports,	

	callback/event-driven architecture, multithreading, P/Invoke interop, Visual Studio 2012, Windows 8/8.1
Role	C++ Platform Integration Developer
Project Overview	Intel-sponsored SDK exposing 2-in-1 convertible hardware state (posture, touch readiness, keyboard availability) to application developers — enabling custom touch-mode UX and suppressing default OS behaviors (e.g., auto on-screen keyboard) without exposing Intel's proprietary driver APIs.
Roles & Responsibilities	• Core developer in a 4-member team; owned the native layer — a C/C++ SDK wrapping Intel's proprietary hardware DLLs behind a stable, versioned C ABI (extern "C" exports, fixed calling conventions, opaque handles to hide internal types). • Implemented a device-state manager normalizing raw hardware signals into posture-transition events (laptop/tablet/tent/presentation) — dispatched via registered callbacks and polling APIs with thread-safe state access and debounced transitions — plus an input-context routing layer enabling applications to programmatically switch input handling (touch vs. keyboard/mouse) per hardware mode. • Hardened managed-to-native interop (P/Invoke callback marshaling, DLL load/unload lifetime, stability under rapid mode transitions); shipped reference adapters and integration docs, validated on Intel Ultrabook-class convertible hardware.

CERTIFICATIONS

- ☐ Microsoft Applied Skills: Enhance agents with autonomous capabilities (2026) — [Verify credential \(Microsoft Learn\)](#)

EDUCATIONAL QUALIFICATIONS

- ☐ Bachelor's in Electronics — Aurora College, 2013

PERSONAL DETAILS

- ☐ Date of Birth: 15 Nov 1992
- ☐ Gender: Male
- ☐ Marital Status: Married
- ☐ Languages Known: English, Telugu, Hindi